Class 4 Task

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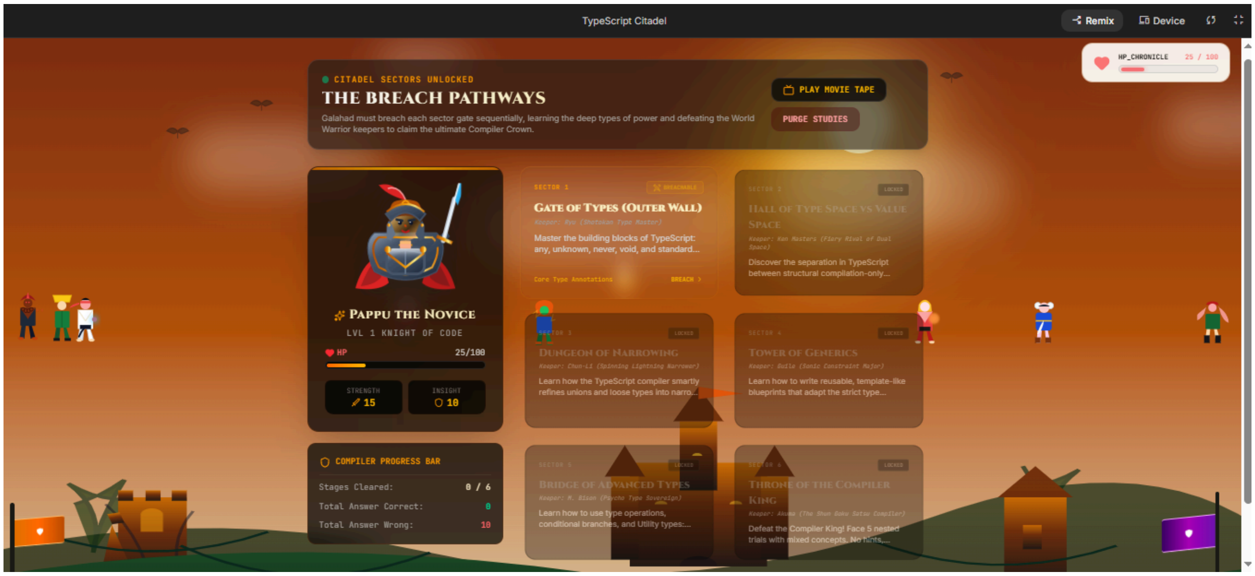
- Project title: TypeScript Citadel
- App/game idea: Learn 6 Core Principles of Typescript
- Purpose of the project: Learn Typescript in a Gamified Quiz way.
- Which AI tools were used: Google AI Studio
- Initial prompt used
- Prompt iterations used during development: Multiple iterations were used to improve the UI and add features.
- What worked well: After adding system prompt it started working better
- What did not work: Without using system prompt it was a bit difficult
- Bugs or challenges faced: Characters descriptions requires precision.
- How the issues were solved: More character description. Using Google AI Studios tools to select the individual div components and then prompt
- Features added or improved: Added Street Fighter Characters, Slides and Movie Feature.
- Final result of the project
- Whether the project solves a real-world problem: Yes it does. It helps a person learn typescript principles in a gamefied way.
- Quiz Game Link: https://ai.studio/apps/5913aca0-c6ad-40ea-8d65-5aaea753dae9

- Screenshot:



- Initial Prompt:

Create an interactive learning game that teaches **6 TypeScript concepts**

The game must:

- Teach concepts through gameplay (not lectures)
- Enforce understanding via challenges
- Adapt difficulty per concept stage
- Provide immediate feedback
- Track user mastery per concept
- 1 Concept at each stage

Game Identity

The player is a **Knight of Code** tasked with conquering the **TypeScript Citadel**.

- The Knight begins at the outer gates of the castle
- Each conquered stage represents mastering a TypeScript concept
- The Knight gains strength, insight, and abilities with each stage
- Final victory = defeating the Compiler King and claiming the throne

1. PLAYER ROLE SYSTEM

1.1 Player Definition

```
player:
role: "Knight of TypeScript"
title_progression:
- "Novice Knight"
- "Gatebreaker Knight"
- "Type Warden Slayer"
- "Generic Vanguard"
- "Royal Code Knight"
- "Master of Knights"

attributes:
knowledge: 0
mastery: 0
resilience: 0

2.1 Castle Identity

castle:
name: "The TypeScript Citadel"
ruler: "The Compiler King"
defense_force:
Create a Fighter against each rule

stages:
```

- stage_1: "Gate of Types (Outer Wall)"
- stage_2: "Hall of Type Space vs Value Space"
- stage_3: "Dungeon of Narrowing"
- stage_4: "Tower of Generics"
- stage_5: "Bridge of Advanced Types"
- stage_6: "Throne of the Compiler King"

```
game_loop:
steps:
1. knight_enters_stage
2. castle_guard_challenge_appears
3. knight_attempts_solution
4. magic_validation_engine_checks
5. outcome_is_revealed
```

```
6. knight_gains_or_loses_strength
7. gate_opens_or_reinforces

tone:
style: "epic medieval fantasy + programming metaphor"

narration_voice:
- "The castle speaks in compiler logic"
- "The Knight thinks in types and structures"
- "Errors appear as cursed runes"

avoid:
- plain technical explanations
- non-immersive tone

stage_rules:
each_stage_must_include:
- knight_entry_narration
- guardian_encounter
- concept_based_challenges
- gate_unlock_mechanism
- narrative_transition

feedback_success:
style: "victory proclamation"
message_elements:
- "The gate collapses before the Knight"
- "The concept bows to mastery"

feedback_failure:
style: "defensive castle reaction"
message_elements:
- "The castle resists the Knight's strike"
- "A rune of confusion appears"
- "The Sage offers a guiding hint"

final_stage:
name: "Throne of the Compiler King"
```

mechanics:
- all_concepts_mixed_trials
- adaptive_difficulty
- no_hints_after_phase_2

win_condition:
- defeat_compiler_king
- pass_all_type_trials

When starting the game:

Spawn player as "Knight of Code"
Place Knight at Outer Gate
Initialize 6 castle stages
Assign first guardian encounter
Begin Stage 1 narration

The game is Basically MCQS Selection game, each concept is tested with 5 MCQs and we reach the next level.
Memory: Use Local Storage
App Type: Only Client Side App
Animations: The app needs to be animation rich
Icons: Need to use battle icons